

MATH3302/MATH7332

Modern cryptanalysis

Feistel ciphers

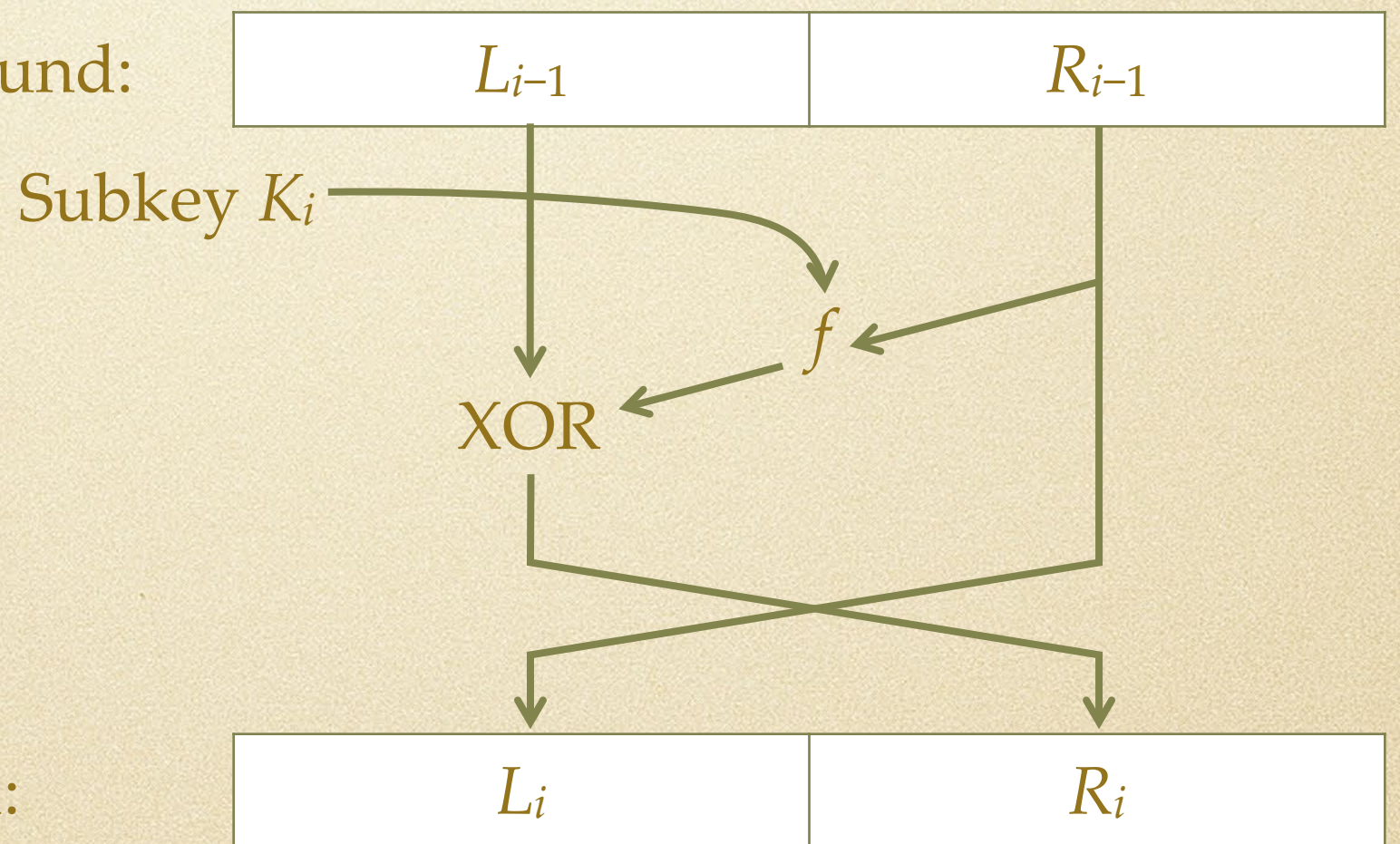
- A broad class of substitution-permutation networks (including the now-decommissioned DES) that are **easy to reverse**
- Encryption and decryption are almost identical
- Involves r **rounds**, each identical except that they use different **subkeys** $K_1, K_2, \dots$
- Subkeys K_i are generated deterministically from the master key K
- Uses a substitution-permutation **function** f :
Input for f : a “half-block” of $(n/2)$ bits, and a subkey K_i
Output for f : a “half-block” of $(n/2)$ bits

Individual rounds

Blocks of **even length**, split into left/right halves

The i th round operates as follows:

Output from $(i-1)$ th round:

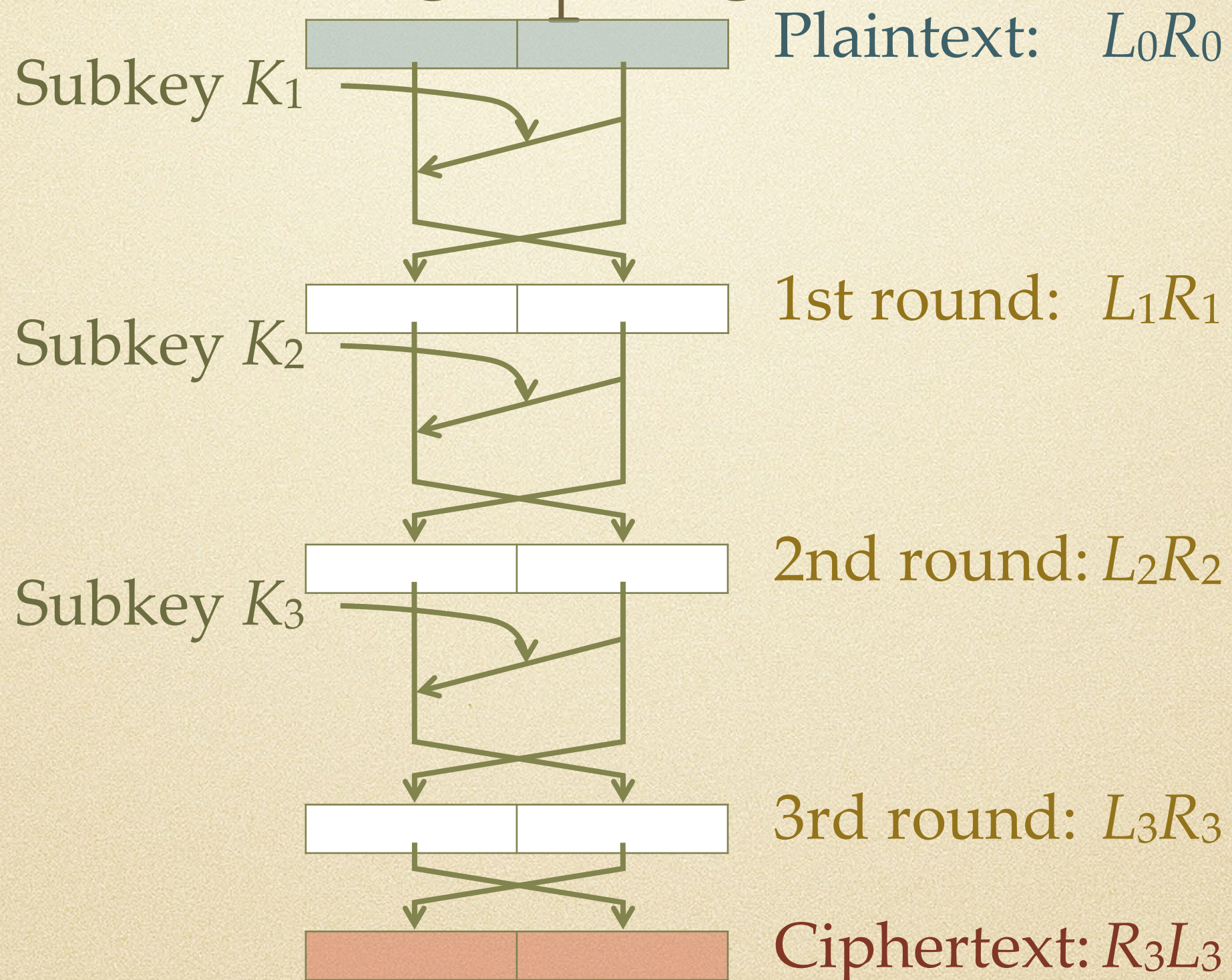


Output from i th round:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$$

A full 3-round cipher



Feistel ciphers

Plaintext: L_0R_0

Round i : $L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$

Ciphertext: R_rL_r

- The function f must be highly non-linear!
This is responsible for the “real” security
- Decryption is identical to encryption,
but uses the subkeys in the reverse order:
 $K_r, K_{r-1}, \dots, K_2, K_1$

DES

- 64-bit blocks, 32-bit half-blocks L_i, R_i
- 56-bit master key K , 48-bit subkeys K_i
- Initial fixed permutation IP , followed by a 16-round Feistel cipher, followed by IP^{-1}
- Each subkey is a permuted subset of 48 bits of K

The DES function f

Recall $R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$ $f: \mathbb{Z}_2^{32} \times \mathbb{Z}_2^{48} \rightarrow \mathbb{Z}_2^{32}$

$f:$

1. Expand R_{i-1} using a fixed “expansion permutation” that repeats 16 elements, making it 48 bits long
2. Add the subkey: XOR with K_i
3. Write the result as $B_1B_2B_3B_4B_5B_6B_7B_8$, where each B_i has 6 bits
4. Substitute each B_i using the S-box S_i , where $S_1 \dots S_8$ are fixed non-linear substitutions that convert 6 bits into 4 bits
5. Apply a final fixed permutation to obtain $f(R_{i-1}, K_i)$

The eight DES S-boxes

- Input bits $b_1b_2b_3b_4b_5b_6$
- Let b_1b_6 denote a row, $b_2b_3b_4b_5$ denote a column
- Read the output bits $c_1c_2c_3c_4$ from a fixed table

S_1 :

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7
1	0	f	7	4	e	2	d	1	a	6	c	b	9	5	3	8
2	4	1	e	8	d	6	2	b	f	c	9	7	3	a	5	0
3	f	c	8	2	4	9	1	7	5	b	3	e	a	0	6	d

Exercise Determine the output for each of the following inputs

(a) 101100

(b) 011000

Designing the S-boxes

- Each row is a permutation of 0123456789abcdef
- Change one input bit $\rightarrow$ changes ≥ 2 output bits
- Not linear or affine functions
- For any input and output positions, fixing the input bit gives $\sim 50\%$ probability of 0 vs 1 in the output bit

S_1 :

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7
1	0	f	7	4	e	2	d	1	a	6	c	b	9	5	3	8
2	4	1	e	8	d	6	2	b	f	c	9	7	3	a	5	0
3	f	c	8	2	4	9	1	7	5	b	3	e	a	0	6	d

Exercise

Input $b_1 b_2 b_3 b_4 b_5 b_6$ Output $c_1 c_2 c_3 c_4$

Fix input bit $b_3 = 0$. Consider the 2^5 possibilities for $b_1 b_2 b_4 b_5 b_6$ and determine how many of those possibilities give output bit $c_1 = 0$.

Linear cryptanalysis

Linear/affine relations involving S-box input and output bits **should** hold with probability $\sim 50\%$

Aims to exploit **biases**, where probabilities differ from 50%

We **combine** linear/affine relations on S-boxes in a way that:

- eliminates bits from intermediate rounds;
- results in biased linear/affine relations involving bits from (a) the plaintext, and (b) the input to the final round

Use a **known plaintext attack** to deduce pieces of the final subkey — essentially, we work out which key bits **exhibit this bias in practice**

Biases

Let X be a random binary variable:

$$\Pr(X = 0) = \frac{1}{2} + \varepsilon, \quad \Pr(X = 1) = \frac{1}{2} - \varepsilon$$

We call ε the **bias** of X .

If X_1 and X_2 are independent binary variables with biases ε_1 and ε_2 , then $X_1 \oplus X_2$ has bias $2\varepsilon_1\varepsilon_2$.

Biases

The piling-up lemma:

If $X_1, X_2, \dots, X_n$ are n independent random binary variables with biases $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$, then $X_1 \oplus \dots \oplus X_n$ has bias $2^{n-1} \prod \varepsilon_i$.

A useful fact:

Suppose X_1, X_2 and X_3 are independent random binary variables, and that $X_1 \oplus X_2$ has bias $\varepsilon_{1,2}$, and $X_2 \oplus X_3$ has bias $\varepsilon_{2,3}$.

Since $X_1 \oplus X_3 = (X_1 \oplus X_2) \oplus (X_2 \oplus X_3)$, and $(X_1 \oplus X_2)$ and $(X_2 \oplus X_3)$ are independent, $X_1 \oplus X_3$ has bias $2\varepsilon_{1,2} \varepsilon_{2,3}$.

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

X_1	X_2	X_3	X_4	Y_1	Y_2	Y_3	Y_4	$X_2 \oplus X_3$	$Y_1 \oplus Y_3 \oplus Y_4$
0	0	0	0	1	1	1	0	0	0
0	0	0	1	0	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0
0	0	1	1	0	0	0	1	1	1
0	1	0	0	0	0	1	0	1	1
0	1	0	1	1	1	1	1	1	1
0	1	1	0	1	0	1	1	0	1
0	1	1	1	1	0	0	0	0	1
1	0	0	0	0	0	1	1	0	0
1	0	0	1	1	0	1	0	0	0
1	0	1	0	0	1	1	0	1	1
1	0	1	1	1	1	0	0	1	1
1	1	0	0	0	1	0	1	1	1
1	1	0	1	1	0	0	1	1	0
1	1	1	0	0	0	0	0	0	0
1	1	1	1	0	1	1	1	0	0

$X_2 \oplus X_3 \oplus Y_1 \oplus Y_3 \oplus Y_4 = 0$ in 12/16 cases $\Rightarrow$ bias = +1/4

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

X_1	X_2	X_3	X_4	Y_1	Y_2	Y_3	Y_4	$X_2 \oplus X_3$	$Y_1 \oplus Y_3 \oplus Y_4$	$X_1 \oplus X_4$	Y_2
0	0	0	0	1	1	1	0	0	0	0	1
0	0	0	1	0	1	0	0	0	0	1	1
0	0	1	0	1	1	0	1	1	0	0	1
0	0	1	1	0	0	0	1	1	1	1	0
0	1	0	0	0	0	1	0	1	1	0	0
0	1	0	1	1	1	1	1	1	1	1	1
0	1	1	0	1	0	1	1	0	1	0	0
0	1	1	1	1	0	0	0	0	1	1	0
1	0	0	0	0	0	1	1	0	0	1	0
1	0	0	1	1	0	1	0	0	0	0	0
1	0	1	0	0	1	1	0	1	1	1	1
1	0	1	1	1	1	0	0	1	1	0	1
1	1	0	0	0	1	0	1	1	1	1	1
1	1	0	1	1	0	0	1	1	0	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	1	1	1	0	0	0	1

$X_1 \oplus X_4 \oplus Y_2 = 0$ in 8/16 cases $\Rightarrow$ bias = 0

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

X_1	X_2	X_3	X_4	Y_1	Y_2	Y_3	Y_4	$X_2 \oplus X_3$	$Y_1 \oplus Y_3 \oplus Y_4$	$X_1 \oplus X_4$	Y_2	$X_3 \oplus X_4$	$Y_1 \oplus Y_4$
0	0	0	0	1	1	1	0	0	0	0	1	0	1
0	0	0	1	0	1	0	0	0	0	1	1	1	0
0	0	1	0	1	1	0	1	1	0	0	1	1	0
0	0	1	1	0	0	0	1	1	1	1	0	0	1
0	1	0	0	0	0	1	0	1	1	0	0	0	0
0	1	0	1	1	1	1	1	1	1	1	1	1	0
0	1	1	0	1	0	1	1	0	1	0	0	1	0
0	1	1	1	1	0	0	0	0	1	1	0	0	1
1	0	0	0	0	0	1	1	0	0	1	0	0	1
1	0	0	1	1	0	1	0	0	0	0	0	1	1
1	0	1	0	0	1	1	0	1	1	1	1	1	0
1	0	1	1	1	1	0	0	1	1	0	1	0	1
1	1	0	0	0	1	0	1	1	1	1	1	0	1
1	1	0	1	1	0	0	1	1	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1	0	1	0
1	1	1	1	0	1	1	1	0	0	0	1	0	1

$$X_3 \oplus X_4 \oplus Y_1 \oplus Y_4 = 0 \text{ in } 2/16 \text{ cases} \Rightarrow \text{bias} = -3/8$$

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

Exercise: Compute
the bias of
 $X_2 \oplus Y_2 \oplus Y_4$.

X_1	X_2	X_3	X_4	Y_1	Y_2	Y_3	Y_4
0	0	0	0	1	1	1	0
0	0	0	1	0	1	0	0
0	0	1	0	1	1	0	1
0	0	1	1	0	0	0	1
0	1	0	0	0	0	1	0
0	1	0	1	1	1	1	1
0	1	1	0	1	0	1	1
0	1	1	1	1	0	0	0
1	0	0	0	0	0	1	1
1	0	0	1	1	0	1	0
1	0	1	0	0	1	1	0
1	0	1	1	1	1	0	0
1	1	0	0	0	1	0	1
1	1	0	1	1	0	0	1
1	1	1	0	0	0	0	0
1	1	1	1	0	1	1	1

Computing all biases

$Y_1 \oplus Y_3 \oplus Y_4$
(1011)

Bias: +4/16

Output terms

Input
terms

$X_2 \oplus X_3$
(0110)

	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	+8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	-2	-2	0	0	-2	+6	+2	+2	0	0	+2	+2	0	0
2	0	0	-2	-2	0	0	-2	-2	0	0	+2	+2	0	0	-6	+2
3	0	0	0	0	0	0	0	0	+2	-6	-2	-2	+2	+2	-2	-2
4	0	+2	0	-2	-2	-4	-2	0	0	-2	0	+2	+2	-4	+2	0
5	0	-2	-2	0	-2	0	+4	+2	-2	0	-4	+2	0	-2	-2	0
6	0	+2	-2	+4	+2	0	0	+2	0	-2	+2	+4	-2	0	0	-2
7	0	-2	0	+2	+2	-4	+2	0	-2	0	+2	0	+4	+2	0	+2
8	0	0	0	0	0	0	0	0	-2	+2	+2	-2	+2	-2	-2	-6
9	0	0	-2	-2	0	0	-2	-2	-4	0	-2	+2	0	+4	+2	-2
a	0	+4	-2	+2	-4	0	+2	-2	+2	+2	0	0	+2	+2	0	0
b	0	+4	0	-4	+4	0	+4	0	0	0	0	0	0	0	0	0
c	0	-2	+4	-2	-2	0	+2	0	+2	0	+2	+4	0	+2	0	-2
d	0	+2	+2	0	-2	+4	0	+2	-4	-2	+2	0	+2	0	0	+2
e	0	+2	+2	0	-2	-4	0	+2	-2	0	0	-2	-4	+2	-2	0
f	0	-2	-4	-2	-2	0	+2	0	0	-2	+4	-2	-2	0	+2	0

Two useful biases

		$Y_2 \quad Y_2 \oplus Y_4$							
		0	1	2	3	4	5	6	7
X_2	0	+8	0	0	0	0	0	0	0
	1	0	0	-2	-2	0	0	-2	+6
	2	0	0	-2	-2	0	0	-2	-2
	3	0	0	0	0	0	0	0	0
	4	0	+2	0	-2	-2	-4	-2	0
	5	0	-2	-2	0	-2	0	+4	+2
	6	0	+2	-2	+4	+2	0	0	+2
	7	0	-2	0	+2	+2	-4	+2	0
$X_1 \oplus X_3 \oplus X_4$	8	0	0	0	0	0	0	0	0
	9	0	0	-2	-2	0	0	-2	-2
	a	0	+4	-2	+2	-4	0	+2	-2
	b	0	+4	0	-4	+4	0	+4	0
	c	0	-2	+4	-2	-2	0	+2	0
	d	0	+2	+2	0	-2	+4	0	+2
	e	0	+2	+2	0	-2	-4	0	+2
	f	0	-2	-4	-2	-2	0	+2	0

$$X_1 \oplus X_3 \oplus X_4 \oplus Y_2 : \text{bias } +1/4$$

$$X_2 \oplus Y_2 \oplus Y_4 : \text{bias } -1/4$$

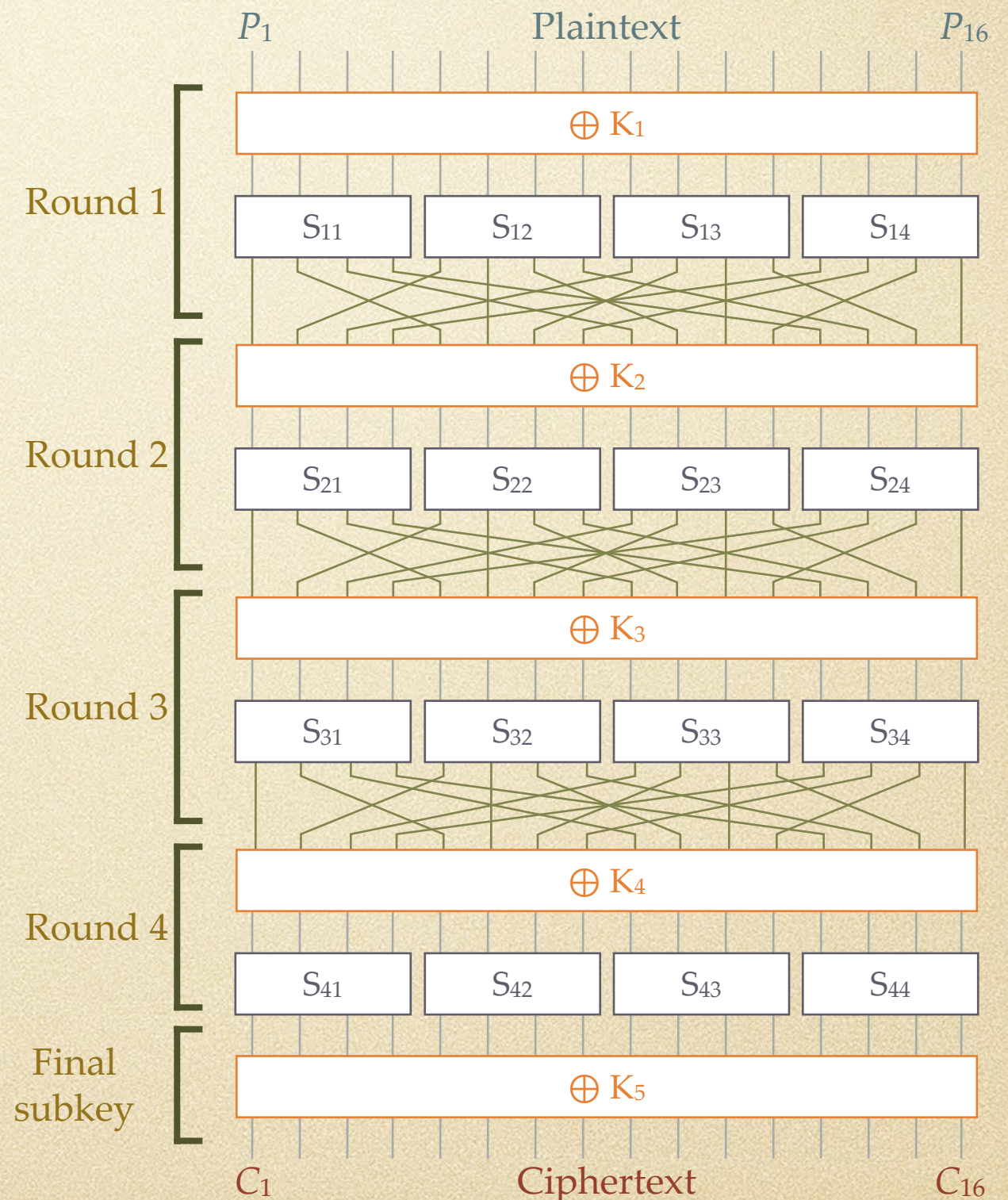
“Useful” means:

- bias is far from 0
- uses few input and output bits

A simple substitution-permutation network

In this example:

- 16-bit blocks
- All S_{ij} are the same S-box just discussed
- Permutation in round 4 would be redundant
- Final subkey K_5 ensures $S_{41} \dots S_{44}$ not redundant

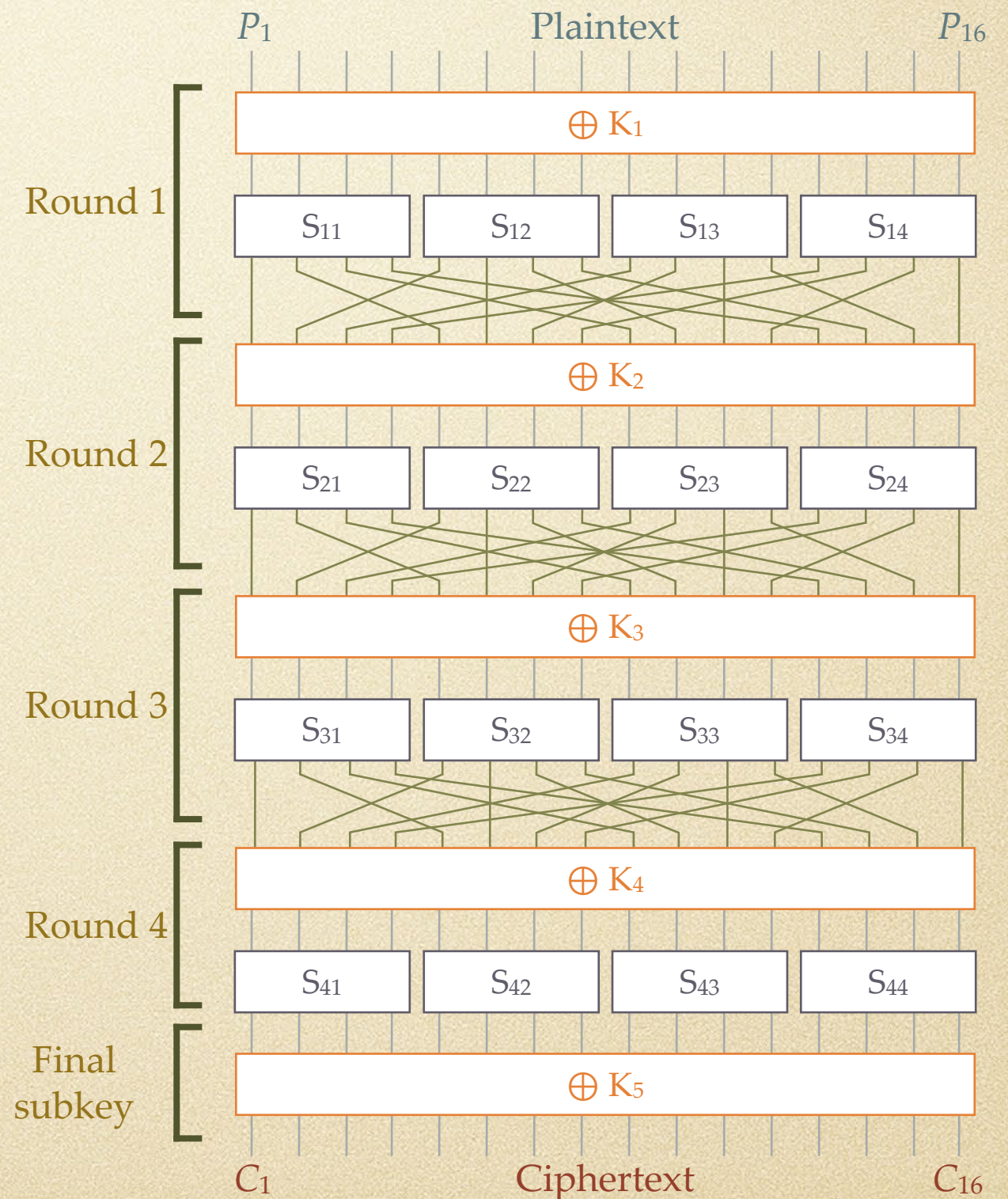


A simple substitution-permutation network

We use the linear relationships with useful biases to follow a few plaintext characters through the system.

$$X_1 \oplus X_3 \oplus X_4 \oplus Y_2 : \text{bias } 1/4$$

$$X_2 \oplus Y_2 \oplus Y_4 : \text{bias } -1/4$$



A simple substitution-permutation network

$$X_1 \oplus X_3 \oplus X_4 \oplus Y_2 : \text{bias } 1/4$$

$$(P_5 \oplus K_{1,5}) \oplus (P_7 \oplus K_{1,7}) \oplus (P_8 \oplus K_{1,8}) \oplus A \Rightarrow \text{bias } 1/4$$

$$X_2 \oplus Y_2 \oplus Y_4 : \text{bias } -1/4$$

$$(A \oplus K_{2,6}) \oplus B \oplus C \Rightarrow \text{bias } -1/4$$

$$(B \oplus K_{3,6}) \oplus D \oplus E \Rightarrow \text{bias } -1/4$$

$$(C \oplus K_{3,14}) \oplus F \oplus G \Rightarrow \text{bias } -1/4$$

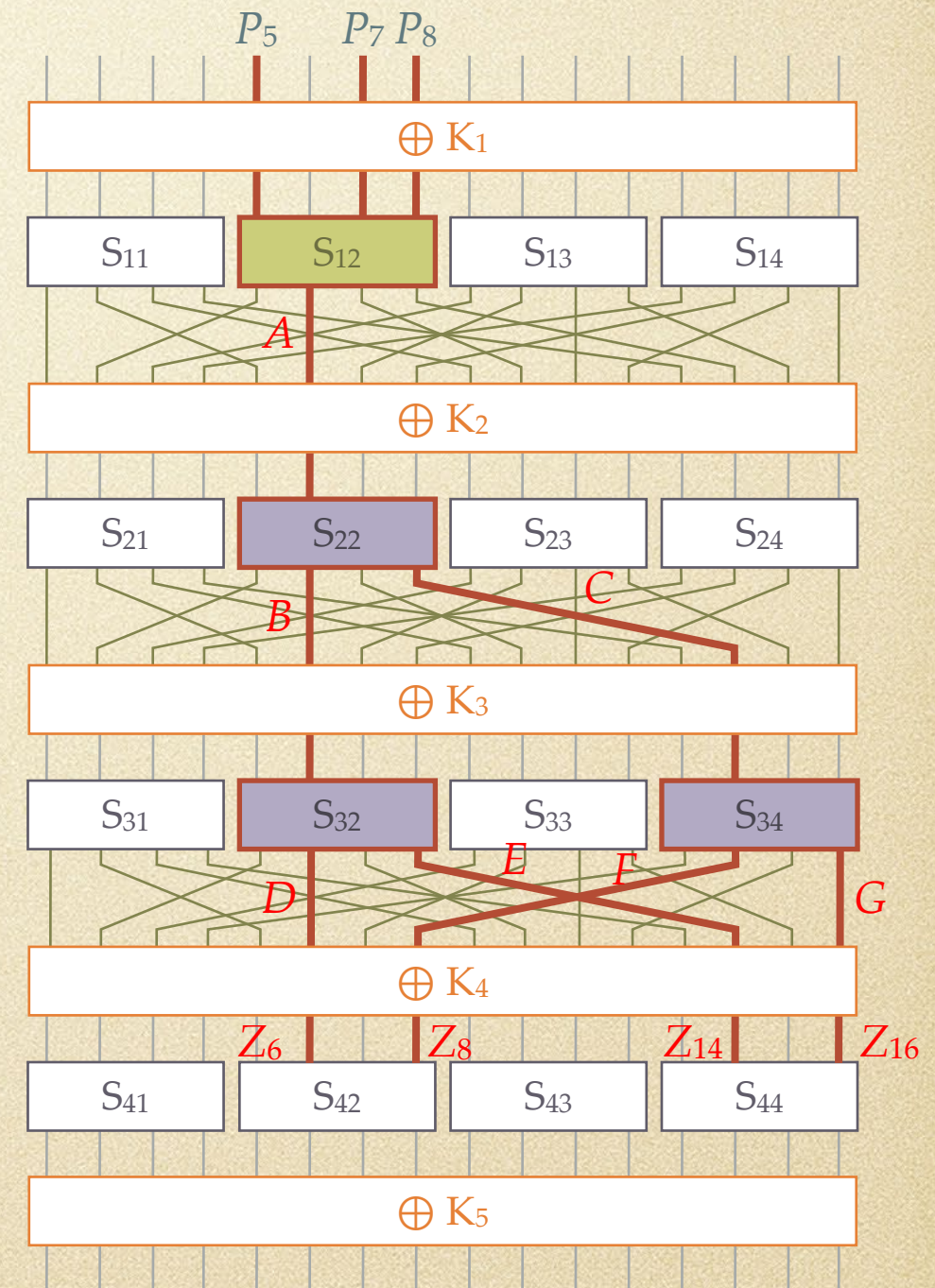
Round 4 key:

$$\begin{array}{ll} Z_6 = D \oplus K_{4,6} & Z_{14} = E \oplus K_{4,14} \\ Z_8 = F \oplus K_{4,8} & Z_{16} = G \oplus K_{4,16} \end{array}$$

Sum, assuming S-boxes are roughly independent:

$$P_5 \oplus P_7 \oplus P_8 \oplus Z_6 \oplus Z_8 \oplus Z_{14} \oplus Z_{16} \\ \oplus \text{some key bits}$$

$$\Rightarrow \text{bias } 2^3 \cdot 1/4 \cdot (-1/4) \cdot (-1/4) \cdot (-1/4) = -0.03125$$



A simple substitution-permutation network

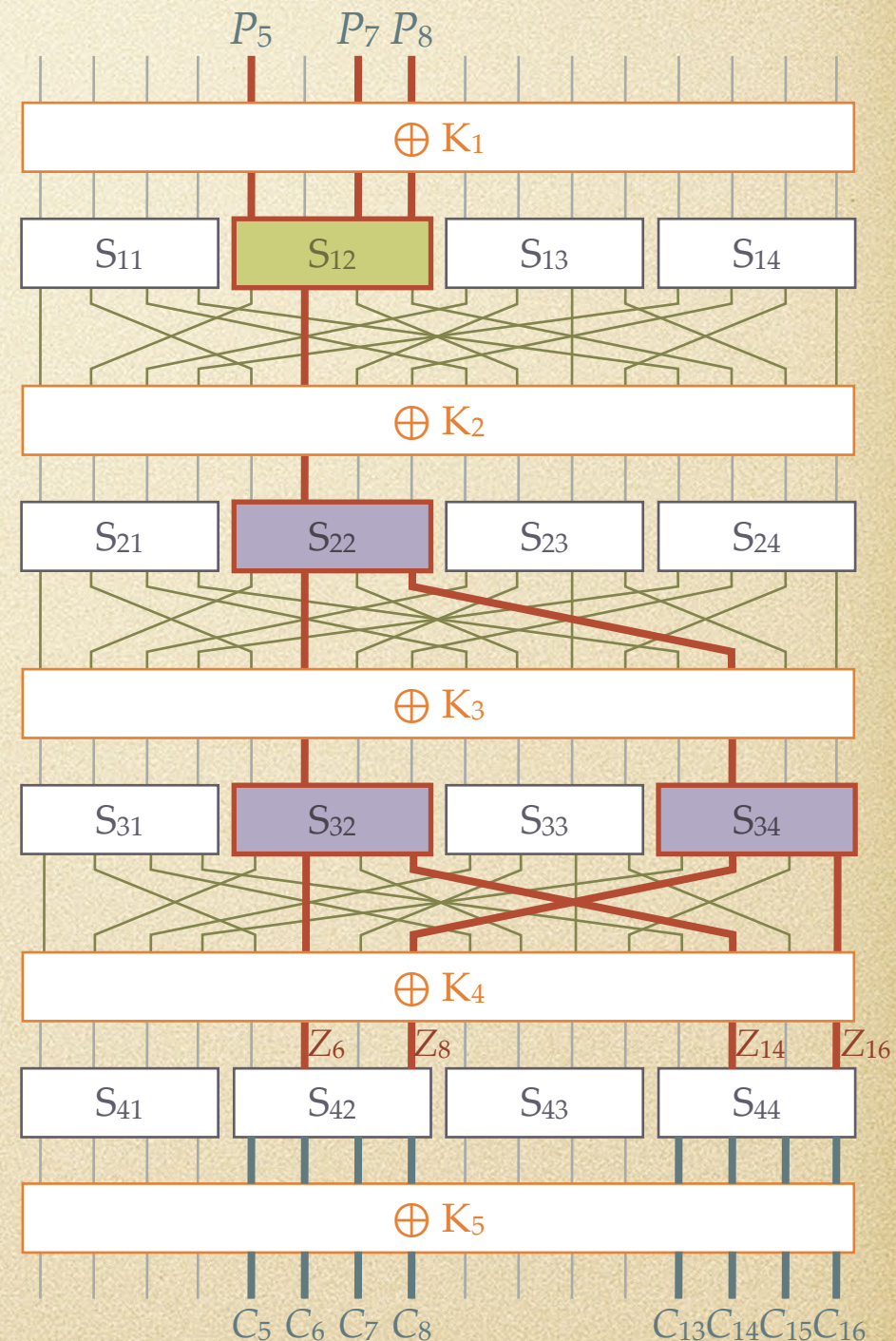
$P_5 \oplus P_7 \oplus P_8 \oplus Z_6 \oplus Z_8 \oplus Z_{14} \oplus Z_{16} \oplus \text{some key bits}$
has bias = -0.03125 .

Therefore:

$P_5 \oplus P_7 \oplus P_8 \oplus Z_6 \oplus Z_8 \oplus Z_{14} \oplus Z_{16}$
must have bias either $+0.03125$ or -0.03125 !

But...

Z_6, Z_8, Z_{14} and Z_{16} are determined purely by
the ciphertext and the final subkey bits
 $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$!



The final step

We know:

- $P_5 \oplus P_7 \oplus P_8 \oplus Z_6 \oplus Z_8 \oplus Z_{14} \oplus Z_{16}$ must have bias either +0.03125 or -0.03125
- Z_6, Z_8, Z_{14} and Z_{16} are determined by the ciphertext and the final subkey bits $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$

So, to launch a **known plaintext attack**:

- Try all 2^8 possibilities for $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$
- Compute the **observed bias** for each, using our known plaintext/ciphertext pairs
- Hope that only one set of key bits exhibits the bias we expect!

Example calculations

Expected bias: ± 0.03125

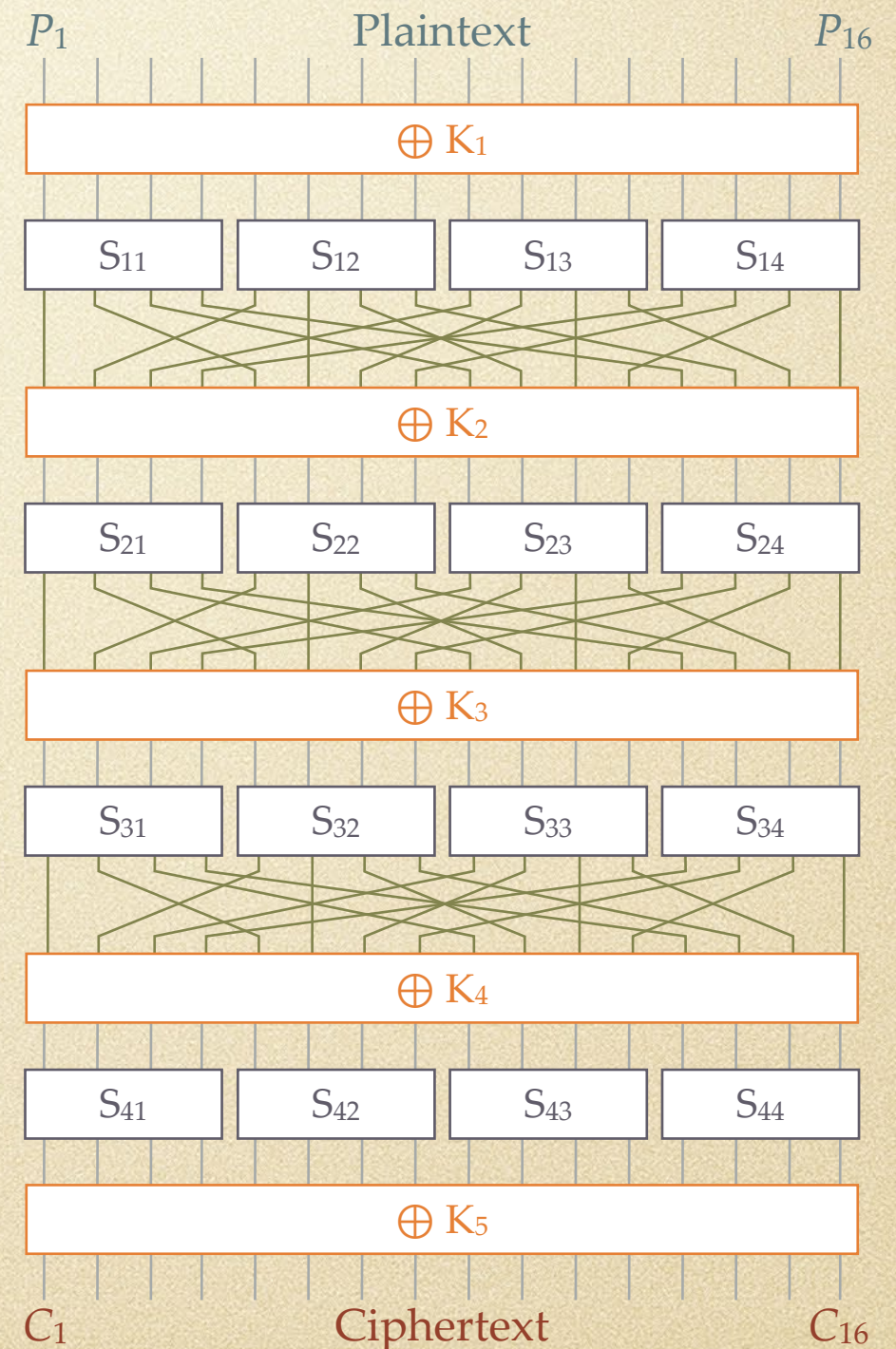
Experiments by Heys on 10,000 plaintext/ciphertext pairs:

Key bits $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$	observed bias
0 0 0 1 1 1 0 0	0.0031
0 0 0 1 1 1 0 1	0.0078
0 0 0 1 1 1 1 0	0.0071
0 0 0 1 1 1 1 1	0.0170
0 0 1 0 0 0 0 0	0.0025
0 0 1 0 0 0 0 1	0.0220
0 0 1 0 0 0 1 0	0.0211
0 0 1 0 0 0 1 1	0.0064
0 0 1 0 0 1 0 0	0.0336
0 0 1 0 0 1 0 1	0.0106
0 0 1 0 0 1 1 0	0.0096
0 0 1 0 0 1 1 1	0.0074
0 0 1 0 1 0 0 0	0.0224
0 0 1 0 1 0 0 1	0.0054

Continuing the attack

In an ideal world:

- Enough of these attacks will expose the **full subkey** K_5
- If K_5 is known, you can deterministically convert from the ciphertext back to the output from $\oplus K_4$
- This effectively **strips one round** from the cipher... and repeat!



How feasible is this?

Let ε be the absolute value of the bias for the final “aggregate” relationship between the $\{ P_i, Z_i \}$ terms

- Number of plaintext/ciphertext pairs required is roughly proportional to $1/\varepsilon^2$
- The bias ε is larger when:
 - the individual S-boxes have larger (absolute) biases
 - fewer S-boxes need to be aggregated together

Good cipher design aims to **minimise the largest such ε**

Differential cryptanalysis

Similar idea to linear cryptanalysis:

- Look for weaknesses in individual S-boxes, then aggregate these across the cipher
- Instead of looking for biased linear/affine relations, we look for biased relationships between ΔX and ΔY (S-box input & output)

Why look at these “differentials” ΔX and ΔY ?

- They do not change when mixing with subkeys!

Unlike linear cryptanalysis, this is a chosen plaintext attack — we need pairs of plaintexts that generate useful differences ΔX

Differentials

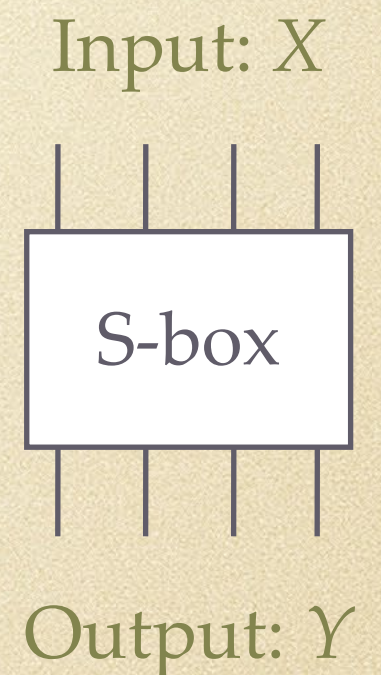
Consider two input/output pairs for an S-box:

- $X' \mapsto Y'$
- $X'' \mapsto Y''$

The input difference is $\Delta X = X' \oplus X''$

The output difference is $\Delta Y = Y' \oplus Y''$

We call the pair $(\Delta X, \Delta Y)$ a differential



Frequencies of differentials

Consider an S-box with n -bit input, n -bit output:

- There are 2^n possible output differences ΔY
- Amongst all pairs of inputs with a fixed input difference ΔX , we might expect each possible ΔY to appear with frequency $\sim 1/2^n$

This is not quite possible, since the input pairs (X', X'') and (X'', X') generate the same output difference ΔY !

However, it is a reasonable rough expectation.

We seek differentials $(\Delta X, \Delta Y)$ that are unusually common: amongst all pairs of inputs with a fixed input difference ΔX , some output difference ΔY occurs much more frequently than expected.

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

Determine the pairs
of inputs that give
 $\Delta X = 1011$

X	Y	Values of ΔY when $\Delta X = 1011$
0 0 0 0	1 1 1 0	
0 0 0 1	0 1 0 0	
0 0 1 0	1 1 0 1	
0 0 1 1	0 0 0 1	
0 1 0 0	0 0 1 0	
0 1 0 1	1 1 1 1	
0 1 1 0	1 0 1 1	
0 1 1 1	1 0 0 0	
1 0 0 0	0 0 1 1	
1 0 0 1	1 0 1 0	
1 0 1 0	0 1 1 0	
1 0 1 1	1 1 0 0	
1 1 0 0	0 1 0 1	
1 1 0 1	1 0 0 1	
1 1 1 0	0 0 0 0	
1 1 1 1	0 1 1 1	

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

Determine the pairs
of inputs that give
 $\Delta X = 1011$

X	Y	Values of ΔY when $\Delta X = 1011$
0 0 0 0	1 1 1 0	0 0 1 0
0 0 0 1	0 1 0 0	0 0 1 0
0 0 1 0	1 1 0 1	0 1 1 1
0 0 1 1	0 0 0 1	0 0 1 0
0 1 0 0	0 0 1 0	0 1 0 1
0 1 0 1	1 1 1 1	1 1 1 1
0 1 1 0	1 0 1 1	0 0 1 0
0 1 1 1	1 0 0 0	1 1 0 1
1 0 0 0	0 0 1 1	0 0 1 0
1 0 0 1	1 0 1 0	0 1 1 1
1 0 1 0	0 1 1 0	0 0 1 0
1 0 1 1	1 1 0 0	0 0 1 0
1 1 0 0	0 1 0 1	1 1 0 1
1 1 0 1	1 0 0 1	0 0 1 0
1 1 1 0	0 0 0 0	1 1 1 1
1 1 1 1	0 1 1 1	0 1 0 1

Differential
 $(\Delta X, \Delta Y) = (1011, 0010)$
 $\Rightarrow$ frequency 8/16

Differential
 $(\Delta X, \Delta Y) = (1011, 0111)$
 $\Rightarrow$ frequency 2/16

Differential
 $(\Delta X, \Delta Y) = (1011, 0101)$
 $\Rightarrow$ frequency 2/16

Differential
 $(\Delta X, \Delta Y) = (1011, 1111)$
 $\Rightarrow$ frequency 2/16

Differential
 $(\Delta X, \Delta Y) = (1011, 1101)$
 $\Rightarrow$ frequency 2/16

Analysing an S-box

4-bit input	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
4-bit output	e	4	d	1	2	f	b	8	3	a	6	c	5	9	0	7

X	Y	Value of $\Delta Y \dots$
		$\dots$ when $\Delta X = 0100$
0 0 0 0	1 1 1 0	
0 0 0 1	0 1 0 0	
0 0 1 0	1 1 0 1	
0 0 1 1	0 0 0 1	
0 1 0 0	0 0 1 0	
0 1 0 1	1 1 1 1	
0 1 1 0	1 0 1 1	
0 1 1 1	1 0 0 0	
1 0 0 0	0 0 1 1	
1 0 0 1	1 0 1 0	
1 0 1 0	0 1 1 0	
1 0 1 1	1 1 0 0	
1 1 0 0	0 1 0 1	
1 1 0 1	1 0 0 1	
1 1 1 0	0 0 0 0	
1 1 1 1	0 1 1 1	

Exercise: Calculate the frequency of the differential $(\Delta X, \Delta Y) = (0100, 0110)$.

All frequencies

Frequency: 8/16

		Output ΔY															
		0010															
Input ΔX		0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
	0	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	0	0	0	2	0	0	0	2	0	2	4	0	4	2	0	0
	2	0	0	0	2	0	6	2	2	0	2	0	0	0	0	2	0
	3	0	0	2	0	2	0	0	0	0	4	2	0	2	0	0	4
	4	0	0	0	2	0	0	6	0	0	2	0	4	2	0	0	0
	5	0	4	0	0	0	2	2	0	0	0	4	0	2	0	0	2
	6	0	0	0	4	0	4	0	0	0	0	0	0	2	2	2	2
	7	0	0	2	2	2	0	2	0	0	2	2	0	0	0	0	4
	8	0	0	0	0	0	0	2	2	0	0	0	4	0	4	2	2
	9	0	2	0	0	2	0	0	4	2	0	2	2	2	0	0	0
	a	0	2	2	0	0	0	0	0	6	0	0	2	0	0	4	0
	b	0	0	8	0	0	2	0	2	0	0	0	0	0	2	0	2
	c	0	2	0	0	2	2	2	0	0	0	0	2	0	6	0	0
	d	0	4	0	0	0	0	0	4	2	0	2	0	2	0	2	0
	e	0	0	2	4	2	0	0	0	6	0	0	0	0	0	2	0
	f	0	2	0	0	6	0	0	0	0	4	0	2	0	0	2	0

Three useful differentials

		0010				0101		0110		
		0	1	2	3	4	5	6	7	
0010	0	16	0	0	0	0	0	0	0	
	1	0	0	0	2	0	0	0	2	
	2	0	0	0	2	0	6	2	2	
	3	0	0	2	0	2	0	0	0	
0100	4	0	0	0	2	0	0	6	0	
	5	0	4	0	0	0	2	2	0	
	6	0	0	0	4	0	4	0	0	
	7	0	0	2	2	2	0	2	0	
1011	8	0	0	0	0	0	0	2	2	
	9	0	2	0	0	2	0	0	4	
	a	0	2	2	0	0	0	0	0	
	b	0	0	8	0	0	2	0	2	
	c	0	2	0	0	2	2	2	0	
	d	0	4	0	0	0	0	0	4	
	e	0	0	2	4	2	0	0	0	
	f	0	2	0	0	6	0	0	0	

(1011, 0010) : freq = 1/2

(0100, 0110) : freq = 3/8

(0010, 0101) : freq = 3/8

“Useful” means:

- frequency is large
- ΔX and ΔY have only a few bits set to 1

Through the network again...

If...

$\Delta = 0000 \ 1011 \ 0000 \ 0000 \longrightarrow$

$\Delta = 0000 \ 1011 \ 0000 \ 0000$ (unchanged by $\oplus K_i$) $\longrightarrow$

$\Delta = 0000 \ 0010 \ 0000 \ 0000$ with freq $\frac{1}{2}$ $\longrightarrow$

$\Delta = 0000 \ 0000 \ 0100 \ 0000$ with freq $\frac{1}{2}$ $\longrightarrow$

$\Delta = 0000 \ 0000 \ 0100 \ 0000$ with freq $\frac{1}{2}$ $\longrightarrow$

$\Delta = 0000 \ 0000 \ 0110 \ 0000$ with freq $\frac{1}{2} \cdot \frac{3}{8}$ $\longrightarrow$

$\Delta = 0000 \ 0010 \ 0010 \ 0000$ with freq $\frac{1}{2} \cdot \frac{3}{8}$ $\longrightarrow$

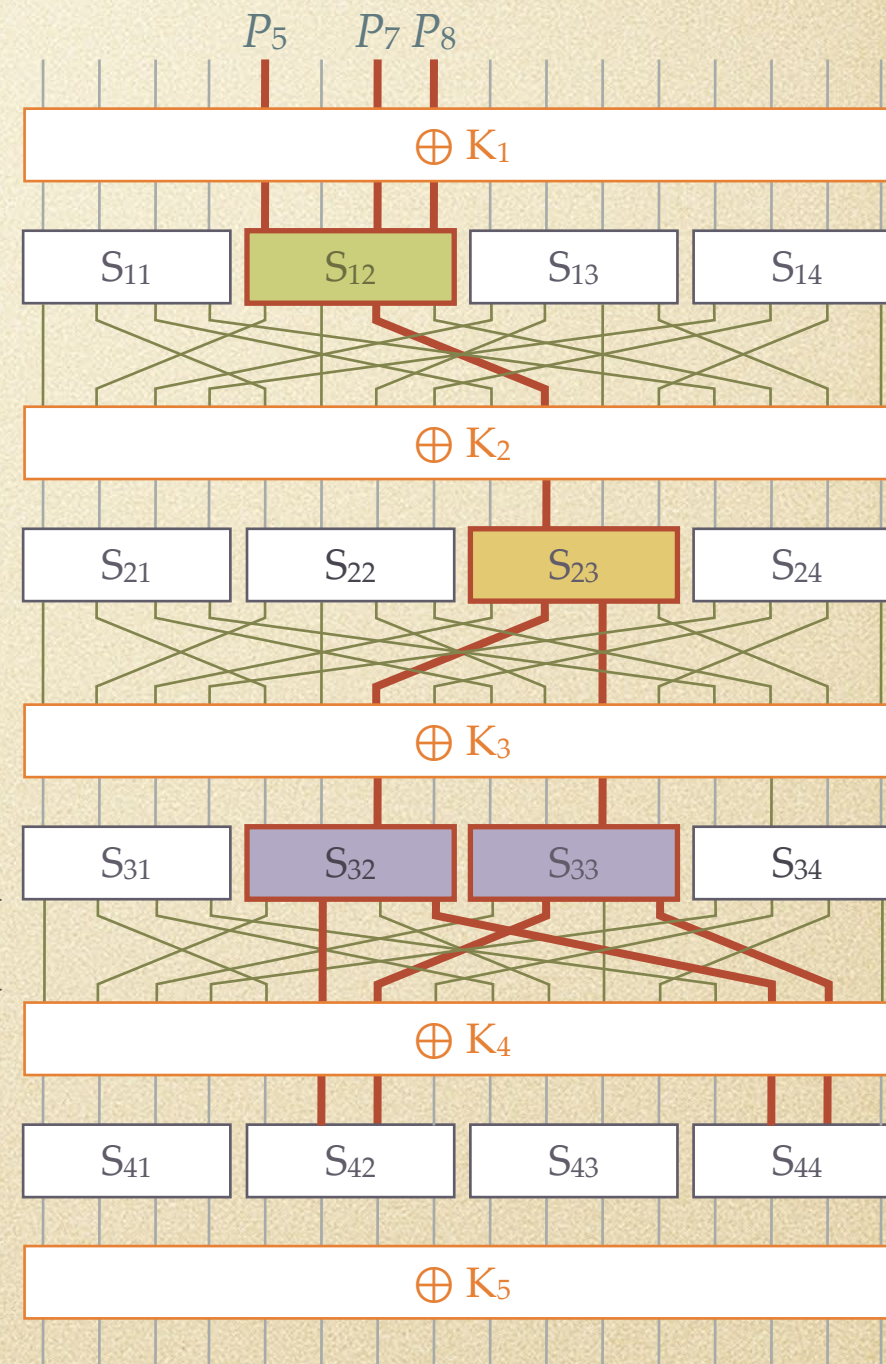
$\Delta = 0000 \ 0010 \ 0010 \ 0000$ with freq $\frac{1}{2} \cdot \frac{3}{8}$ $\longrightarrow$

$\Delta = 0000 \ 0101 \ 0101 \ 0000$ with freq $\frac{1}{2} \cdot \frac{3}{8} \cdot \frac{3}{8} \cdot \frac{3}{8}$ $\longrightarrow$

$\Delta = 0000 \ 0110 \ 0000 \ 0110$ with freq $\frac{1}{2} \cdot \frac{3}{8} \cdot \frac{3}{8} \cdot \frac{3}{8}$ $\longrightarrow$

$\Delta = 0000 \ 0110 \ 0000 \ 0110$ with freq $\frac{1}{2} \cdot \frac{3}{8} \cdot \frac{3}{8} \cdot \frac{3}{8}$ $\longrightarrow$

Again we assumed S-boxes are
roughly **independent**



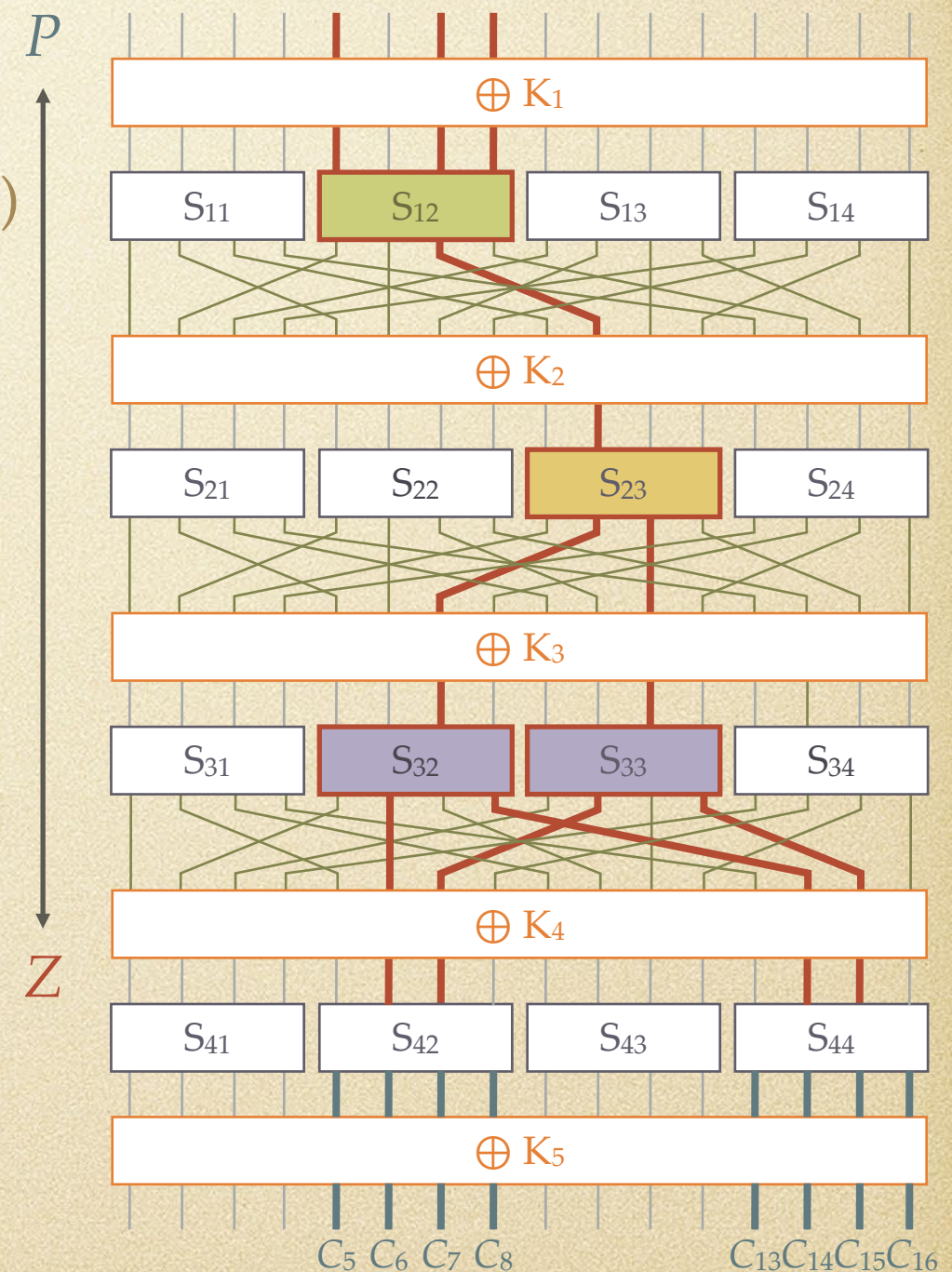
Our final differential

$$(\Delta P, \Delta Z) = (0000 \ 1011 \ 0000 \ 0000, 0000 \ 0110 \ 0000 \ 0110)$$

If we fix this input difference ΔP ,
then this output difference ΔZ has frequency
 $\sim 1/2 \cdot 3/8 \cdot 3/8 \cdot 3/8 = 27/1024$

But once again...

Z is determined purely by
the ciphertext and the final subkey bits
 $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$!



The final step

We know:

- If we fix $\Delta P = 0000\ 1011\ 0000\ 0000$, then $\Delta Z = 0000\ 0110\ 0000\ 0110$ should occur with frequency $\sim 27/1024$
- Z is determined by the ciphertext and the final subkey bits $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$

So, to launch a **chosen plaintext attack**:

- **Generate many plaintext pairs** with input difference $\Delta P = 0000\ 1011\ 0000\ 0000$
- Try all 2^8 possibilities for $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$
- For each possibility, measure the **observed frequency** of the resulting difference $\Delta Z = 0000\ 0110\ 0000\ 0110$
- Hope that only one set of key bits exhibits the frequency we expect!

Example calculations

Expected frequency: $27/1024 \sim 0.0264$

Experiments by Heys on 5,000 pairs of plaintexts:

Key bits $K_{5,5}, K_{5,6}, K_{5,7}, K_{5,8}, K_{5,13}, K_{5,14}, K_{5,15}, K_{5,16}$	Observed frequency
0 0 0 1 1 1 0 0	0.0000
0 0 0 1 1 1 0 1	0.0000
0 0 0 1 1 1 1 0	0.0000
0 0 0 1 1 1 1 1	0.0000
0 0 1 0 0 0 0 0	0.0000
0 0 1 0 0 0 0 1	0.0136
0 0 1 0 0 0 1 0	0.0068
0 0 1 0 0 0 1 1	0.0068
0 0 1 0 0 1 0 0	0.0244
0 0 1 0 0 1 0 1	0.0000
0 0 1 0 0 1 1 0	0.0068
0 0 1 0 0 1 1 1	0.0068
0 0 1 0 1 0 0 0	0.0030
0 0 1 0 1 0 0 1	0.0024

Feasibility

Let f be the “aggregate” expected frequency for our $(\Delta P, \Delta Z)$ differential, assuming ΔP is fixed

- Number of plaintext pairs we need to generate is roughly proportional to $1/f$
- This frequency f is larger when:
 - the individual S-boxes have larger differential frequencies
 - fewer S-boxes need to be aggregated together

Again, good cipher design aims to **minimise the largest such f**